
Revision Table

Revision No.	Details	Section / Page No.
1	Initial thesis structure setup	All chapters
2	Added LaTeX packages for code and formatting	Preamble
3	Restructured to match approved ClassiFish format	Chapter organization
4	[Future revisions to be added here]	[Location]

Table 1: Thesis Revision History



Consol: A Notes-to-Recollection Application Using SimCSE for its Textual Semantic Comparison

George Christopher C. Daguman

Supervised by
Dante D. Dinawanao

Department of Computer Science
College of Computer Studies
MSU - Iligan Institute of Technology

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Abstract

Generative AI has presented society with an opportunity to create a surplus of educational content in mere seconds. While knowledge is inherently beneficial and promotes intellectual and academic growth, there still is a bridge to actual learning that cannot be crossed just through using AI and its instantaneous generative abilities (Mishra & Anthony, 2023). Learning, in the academic landscape, is comprised of obtaining information which can be in the form of notes (Rashid & Rigas, 2006; Devlin et al., 2019). Retaining knowledge after note-taking remains a challenge, as forgetting is inevitable and overlooked by recent AI and NLP advancements. (Klemm, 2012; Naim et al., 2020). In studying and reviewing, if a person wishes to ensure their recollection of key topics is accurate and coherent, options are limited without manually referencing source material, which can be tiring and inefficient. Assistance is needed, particularly in providing instant feedback for accuracy and mastery of notes (Karpicke & Roediger, 2008; Sachs, 1967). This research proposes Consol, a notes-to-recollection application that uses SimCSE's contrastive sentence embeddings and semantic similarity analysis to compare how similar a person's recollection is to their original notes (Gao et al., 2021; Corley et al., 2005), and uses progress tracking techniques to encourage consistency and improvement (Deterding et al., 2011; Gaston & Cooper, 2017). It will be a web-based application that leverages semantic similarity evaluation to assess memory recall effectively. The study aims to evaluate whether semantic-based feedback enhances learning outcomes compared to traditional recall methods. Through system testing and user performance assessment, this research will analyze the system's usability, accuracy, and effectiveness in improving memory retention (Baars et al., 2022; Tran & Nguyen, 2022). The results aim to provide a scalable solution bridging note-taking and active learning while fostering consistent knowledge consolidation (Vişcu, 2024; Stojanovic et al., 2020).

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Introduction

Memory is not merely the result of recording information but the outcome of consistent review and active engagement with what has been learned. Notes, while essential for capturing ideas, do not inherently lead to long-term retention or understanding simply by being created. Instead, it is through revisiting, processing, and consolidating these notes that individuals truly reinforce their knowledge (Karpicke and Roediger, 2008; Klemm, 2012).

For students, teachers, and professionals alike, the challenge lies in establishing consistent learning practices and receiving validation for their efforts—validation that not only measures recall but also provides meaningful feedback on comprehension (Stojanovic et al., 2020; Baars et al., 2022). Traditional note-taking methods often focus on transcription, prioritizing the exact wording of information over its deeper meaning. However, research by Sachs (1967) highlights that individuals naturally retain the semantic meaning—the "gist" of information—rather than its verbatim form.

This tendency to paraphrase and restructure ideas reflects the cognitive process of abstraction, where understanding takes precedence over memorization. Conventional systems that evaluate recall by word-for-word accuracy fail to account for this natural process, offering an incomplete and often discouraging measure of learning performance (Corley et al., 2005; Gao et al., 2021).

Additionally, progress tracking has proven to be an effective strategy for sustaining motivation and fostering consistent learning habits. By incorporating elements such as performance metrics, mastery indicators, and progress tracking, such systems provide users with actionable insights and a sense of achievement (Deterding et al., 2011; Gaston and Cooper, 2017). These tools not only validate the user's learning progress but also

make the process more interactive and rewarding, promoting deeper engagement and long-term retention (Tran and Nguyen, 2022; Viscu, 2024).

1.1 | Background of the Study

Now, as technology continues to advance in the current “information age,” the demand for fast, accurate, and easily accessible information has increased dramatically. This is particularly evident in fields where knowledge acquisition is critical, such as education, research, and personal development. The proliferation of generative AI technologies and digital tools has revolutionized the ways individuals gather, process, and manage information. Digital note-taking tools, such as Evernote, Notion, Obsidian, and various AI-assisted applications, offer features like automatic transcription, cloud synchronization, and summarization, making the recording of ideas and knowledge more efficient and convenient to the modern user (Baars et al., 2022; Rashid and Rigas, 2006).

Existing studies confirm that effective learning requires more than passive note creation. Research conducted by Sachs (1967) indicates that individuals are better at retaining semantic meaning rather than exact syntactic details. This means that during memory retrieval, individuals often paraphrase, restructure, and abstract ideas rather than reproduce content verbatim. Traditional systems that rely on word-for-word accuracy to evaluate recall fail to account for this natural cognitive process, leading to an incomplete understanding of memory performance (Corley et al., 2005; Gao et al., 2021).

Building on this, the integration of AI-driven systems in learning environments has opened new avenues for assessing memory and comprehension more effectively. By including the more recent advancements in natural language processing (NLP) and similarity-based algorithms, these systems can evaluate recall beyond mere verbatim reproduction, focusing instead on the semantic alignment between original and recalled content (Devlin et al., 2019; Gao et al., 2021). Such approaches recognize the inherent flexibility of human cognition, where meaningful understanding takes precedence over exact replication. For example, sentence-transformer models like SimCSE have demonstrated the ability to measure semantic similarity with high accuracy, making them suitable for applications that assess memory recall in a way that mirrors human evaluation (Gao et al., 2021; Corley et al., 2005).

This shift toward semantic-based assessment not only provides a more accurate

measure of knowledge retention but also aligns with modern pedagogical principles that emphasize deep learning and critical thinking. By rewarding comprehension and conceptual mastery over rote memorization, AI-assisted systems encourage learners to engage more actively with the material, fostering long-term knowledge retention and improved cognitive performance (Karpicke and Roediger, 2008; Klemm, 2012). As these technologies continue to evolve, they hold the potential to transform traditional methods of learning evaluation and pave the way for more adaptive and personalized educational tools (Stojanovic et al., 2020; Viscu, 2024).

1.2 | Statement of the Problem

The rapid development and advancement of Generative AI has led to the overwhelming abundance of knowledge, and with that, an unlimited opportunity to take notes. While this is clearly for some benefit, it is not put into actual use if the increasing number of notes are not consolidated into memory, thereby hindering the cumulative learning process (Karpicke & Roediger, 2008; Klemm, 2012). Additionally, in individual study contexts, there is a lack of mechanisms to verify whether a learner has truly understood or mastered the material. This problem is compounded by the fact that human recall often involves paraphrasing and abstracting information, making it difficult for current systems to accurately assess a learner's understanding (Sachs, 1967; Corley et al., 2005). Consequently, there is a need for a system that can evaluate learning by recognizing and comparing both the original content and its reconstructed ideas through context clues, ensuring that abstracted yet relevant concepts are acknowledged without overlooking their presence in the material (Gao et al., 2021; Devlin et al., 2019).

1.3 | Objectives

1.3.1 | General Objective

To develop a web-based memorization system that refers to the user's pre-made notes and prompts the user to perform active textual recall, leveraging SimCSE to maintain fairness during comparison despite non-verbatim phrasing of ideas.

1.3.2 | Specific Objectives

1. Design the system architecture of the notes-to-recollection comparison system, including its scoring criteria for evaluating recall accuracy.
2. Devise pre-processing procedures for removing unnecessary non-alphanumeric symbols and formatting artifacts to be suitable for semantic comparison.
3. Employ Simple Contrastive Sentence Embeddings (SimCSE) for semantic comparison of two given texts.
4. Implement a web-based application integrating the SimCSE Model, giving it an appropriate user experience that facilitates motivation through progress tracking and active learning.
5. Conduct a usability test on the web-based application and document the results.

1.4 | Scope and Limitation

The study is strictly limited to a maximum of 10 test participants. The study will be conducted from January 2025 through April of the same year.

1.5 | Significance and Contribution

The proposed implementation will assist users in memorizing their notes and incentivize steady progress in content mastery through various tracking strategies.

This research contributes to the field through several key innovations: the first educational application of SimCSE contrastive learning for semantic similarity assessment, demonstrating novel integration of unsupervised contrastive learning principles with educational technology; development of a multi-dimensional assessment framework that evaluates recall through semantic similarity, completion metrics, and temporal performance indicators; implementation of memory consolidation theory through practical digital application, bridging cognitive science principles with educational technology; and advancement of objective educational assessment by reducing subjective bias through consistent semantic evaluation.

The system's name "Consol" reflects the fundamental educational principle of memory consolidation, where the brain naturally compresses and abstracts information

for long-term storage. Rather than penalizing students for paraphrasing or conceptual restructuring, the system recognizes that consolidation inherently involves "compaction of bulk data," allowing semantic understanding to take precedence over verbatim reproduction. This approach aligns with natural cognitive processes while maintaining academic rigor through objective similarity measurement.

The research addresses critical gaps in educational technology where traditional assessment methods fail to account for semantic equivalence, providing a foundation for future adaptive learning systems that recognize and reward genuine comprehension over superficial memorization.

Review of Related Literature

This chapter reviews relevant literature on note-taking strategies, memory processes, natural language processing (NLP) for semantic similarity, and progress tracking techniques. These areas provide the theoretical basis for understanding information encoding, memory consolidation, textual comparison, and user engagement. By examining these interconnected concepts, the review establishes a foundation for exploring methods that enhance learning, recall, and performance. The studies discussed offer insights into optimizing knowledge retention, evaluating recall accuracy, and motivating users through structured feedback mechanisms.

2.1 | Memory and Cognitive Processes

Memory plays a crucial role in the learning process, specifically in the retention and recall of information. Studies have demonstrated that memory prioritizes the retention of meaning over surface-level details, which has direct implications for comprehension and learning.

2.1.1 | Working Memory and Learning Efficiency

Klemm (2007) establishes working memory as the cognitive foundation for intellectual performance, functioning as a "scratch pad" where information chunks are processed during thinking operations. The tight correlations between working memory capacity, IQ, and problem-solving abilities demonstrate that students' learning effectiveness depends on their ability to maintain and manipulate information during cognitive tasks, challenging contemporary educational approaches that emphasize understanding while neglecting memorization skills.

This research reveals that robust memory expands cognitive capabilities and expedites new understanding development. Students cannot effectively apply concepts they understand if they cannot remember them when needed, establishing memory as essential for both knowledge retention and intellectual development rather than mere rote learning.

2.1.2 | The Role of Semantic and Syntactic Memory

Sachs (1967) explored the relationship between semantic and syntactic memory in the context of comprehension. In her study, participants were presented with sentences and tested on their ability to recognize changes in meaning and form after brief intervals. The findings revealed that memory retention is primarily focused on semantic content, while syntactic details, such as active-to-passive transformations, fade quickly. Even when participants retained the overall meaning of the sentences, the formal properties of the sentences were quickly forgotten. This study concludes that memory is structured to prioritize the "deep structure" or meaning of a sentence over its "surface structure" or syntax, suggesting that comprehension relies on extracting and retaining the core meaning of information.

Sachs' temporal dynamics research reveals that after 80-160 syllables of interpolated material, recognition for syntactic changes dropped to near chance levels while semantic recognition remained robust, demonstrating that individuals naturally retain meaning rather than exact form. This natural tendency toward semantic retention over syntactic precision provides theoretical justification for similarity-based evaluation systems that align with natural memory processes, supporting assessment approaches that recognize semantic equivalence while maintaining academic rigor.

2.1.3 | Cognitive Processes in Memory Retention

Rashid and Rigas (2006) provide an analysis of memory stages and their relationship to learning and information retention. The study outlines the progression of information through sensory memory, short-term memory, and long-term memory, highlighting the necessity of active cognitive processing for effective retention. Encoding, which involves the initial capture of relevant information, and reviewing, which reinforces the transfer of information to long-term memory, are identified as critical phases in the learning process. This study also discusses retrieval failure as a cause of forgetting, where proactive interference and retroactive interference disrupt recall.

Proactive interference occurs when previously learned information impairs the learning of new material, while retroactive interference arises when newly acquired knowledge interferes with the recall of older information. These findings emphasize that memory retention is optimized when learners engage in deep processing activities, such as organizing and summarizing information, rather than relying on surface-level strategies. This perspective aligns with the findings of Sachs (1967), reinforcing the importance of meaningful encoding and the prioritization of core content over superficial details.

2.1.4 | Memory and Recall

Naim et al. (2020) identified that recall performance grows sublinearly as the list length increases. This sublinear growth highlights a saturation effect, where the number of items retrieved plateaus despite an increase in encoded material. The study also revealed that encoding quality plays a crucial role in determining recall performance. Slower presentation rates allowed participants to encode a higher number of words into memory, which improved the total recall capacity. These findings provide evidence that memory recall adheres to universal laws, offering a structured and deterministic perspective on retrieval processes.

2.1.4.1 | Retrieval-Based Testing

Karpicke and Roediger (2008) emphasize that testing is not merely a means of assessment but a potent strategy for enhancing long-term memory retention. Their study debunks the conventional belief that once information is initially learned, further practice adds little value. Instead, repeated retrieval—through testing—promotes deeper encoding and strengthens memory consolidation, highlighting retrieval as an active mechanism essential for sustained learning performance.

2.1.4.2 | Optimizing Retrieval Practice for Long-term Retention

Karpicke and Roediger (2008) demonstrated that retrieval practice plays a central role in consolidating memories and improving long-term retention. Their experiments revealed that repeated retrieval significantly outperforms repeated study in enhancing memory performance. This finding validates the importance of active recall over passive encoding strategies.

The systematic investigation by Karpicke and Roediger reveals critical distinctions between study repetition and test repetition effects that challenge conventional

educational assumptions. Once students correctly produce learning items, continued studying provides no measurable retention benefit, whereas continued testing generates substantial positive effects on long-term memory consolidation. This differential impact suggests that retrieval practice engages cognitive mechanisms distinct from those involved in passive review.

Furthermore, students' predictions of their own learning performance show no correlation with actual retention outcomes, indicating poor metacognitive awareness of effective learning strategies. This metacognitive blindness suggests that students may persist with ineffective study methods while avoiding retrieval practice that would substantially improve their learning outcomes. These findings provide empirical support for educational systems that prioritize active recall over passive content review.

2.1.5 | Mathematical Models of Memory Recall

Naim et al. (2020) demonstrate that human memory recall, despite appearing fragile and unpredictable, follows universal mathematical laws that can be derived from first principles. Their research on free recall paradigms reveals that memory operates as a deterministic process rather than a random phenomenon, with recall performance following predictable mathematical relationships.

The fundamental law of memory recall establishes that the average number of items recalled (R) from M items in memory follows the relationship $R = \sqrt{\frac{3\pi M}{2}}$. This parameter-free prediction emerges from modeling memory as associative search processes operating on random graphs defined by memory representation structure.

The theoretical foundation rests on two principles: memory items are represented by overlapping random sparse neuronal ensembles in dedicated memory networks, and recall follows associative chains where the next item recalled has the largest overlap with the current item. This associative retrieval mechanism provides mathematical validation for similarity-based recall processes, supporting computational approaches to memory assessment that rely on semantic similarity rather than exact matching.

2.2 | Note-Taking Techniques and Tools

Note-taking serves as an essential tool for facilitating memory retention and comprehension. Various techniques and tools have been developed to assist learners in organizing and processing information effectively.

2.2.1 | Traditional Note-Taking Techniques

Rashid and Rigas (2006) examine the comparative effectiveness of traditional note-taking methods, including the Cornell Method, the Outlining Method, and the Mapping Method. These methods provide learners with strategies for organizing content to improve comprehension and retention.

The Cornell note-taking method consists of three distinct sections: keywords area, notes area, and summary area, providing systematic organization without multiple recopying. Rashid and Rigas demonstrate its effectiveness across diverse subject domains through active interaction requirements that enable immediate review and improved comprehension via individual understanding during summarization processes. While alternative methods like outlining provide hierarchical organization, they require intensive concentration unsuitable for fast-paced lectures, and mapping methods demand active participation that may not be feasible in all educational contexts.

2.2.2 | Digital Note-Taking and E-Learning Tools

Rashid and Rigas (2006) further explore the role of technology in enhancing note-taking and memory retention within e-learning environments. The integration of tools such as Microsoft OneNote and Tablet PCs has provided learners with new ways to engage with content, enabling multimedia annotations and portable accessibility.

Rashid and Rigas identify seven critical parameters for effective e-learning environments: institutional support, course development, teaching integration, course structure, student support, faculty support, and evaluation frameworks. Digital note-taking technologies enable knowledge consolidation through multimedia integration where students combine textual notes with audio, visual, and interactive elements, supporting encoding phases of memory formation while providing searchable, editable, and shareable content.

The effectiveness depends on students' learning style classifications: divergers, assimilators, convergers, and accommodators, each requiring different technological approaches. The research demonstrates that effective digital tools enhance motivation while supporting academic achievement through appropriate technological integration that follows established learning models.

2.3 | Natural Language Processing for Semantic Similarity

Semantic similarity measurement represents a fundamental challenge in natural language processing, where the goal is to quantify the degree of meaning correspondence between textual units. This capability serves as the foundation for automated assessment systems that must evaluate student responses based on conceptual understanding rather than superficial text matching.

2.3.1 | Evolution of Semantic Similarity Approaches

Corley, Csomai, and Mihalcea (2005) establish the theoretical foundations for computational semantic similarity measurement, demonstrating that semantic relationships can be quantified through lexical analysis, semantic networks, and corpus-based statistical approaches. Their work identifies critical limitations in traditional approaches that rely primarily on lexical overlap, which fails to capture semantic relationships between synonymous expressions and conceptually equivalent phrasings.

The evolution from lexical to semantic approaches represents a fundamental shift in natural language understanding, where surface-level text matching gives way to meaning-based comparison. Corley et al. demonstrate that semantic similarity measurement must account for paraphrasing, conceptual equivalence, and contextual meaning variation to provide educationally meaningful assessment capabilities.

2.3.2 | Transformer Architecture and Contextualized Embeddings

Devlin et al. (2019) introduce BERT (Bidirectional Encoder Representations from Transformers), establishing the theoretical and practical foundations for contextualized language understanding. BERT's bidirectional training enables the model to consider both left and right context when generating word representations, addressing fundamental limitations of previous approaches that processed text unidirectionally.

The BERT architecture employs attention mechanisms that enable the model to selectively focus on relevant contextual information when generating representations. This attention-based approach proves particularly valuable for educational applications where word meaning depends critically on surrounding context. The model's ability to capture syntactic and semantic relationships through self-supervised learning

eliminates the need for labeled similarity datasets while providing robust performance across diverse domains.

2.3.3 | Contrastive Learning for Sentence Embeddings

Gao, Yao, and Chen (2021) address critical limitations in applying pre-trained language models to sentence similarity tasks through Simple Contrastive learning of Sentence Embeddings (SimCSE). Their research reveals that standard BERT embeddings suffer from anisotropic distribution problems where sentence representations cluster in narrow regions of the vector space, artificially inflating similarity scores regardless of actual semantic relationships.

SimCSE solves this fundamental problem through contrastive learning that encourages semantically similar sentences to have similar embeddings while pushing dissimilar sentences apart. The key innovation involves using dropout as a data augmentation mechanism, where the same sentence processed with different dropout masks serves as positive pairs for contrastive training. This approach proves particularly valuable for educational applications where assessment systems must distinguish between genuine semantic similarity and superficial text overlap.

2.3.3.1 | Quizlet

Tran and Nguyen (2022) demonstrate that Quizlet integrates notes, spaced repetition, and active learning into an engaging platform, effectively improving vocabulary retention and long-term memory consolidation.

Tran and Nguyen's action research reveals that Quizlet application produces significant positive effects on students' vocabulary retention through systematic integration of short-term and long-term memory processes. The platform addresses critical gaps in traditional vocabulary instruction where students easily forget new words without extended practice, particularly in contexts where curriculum time constraints limit intensive vocabulary development.

The effectiveness of Quizlet stems from its implementation of word retention principles, defined as the ability to provide meaning of new words after extended time periods. The application facilitates knowledge consolidation by enabling students to engage with vocabulary both in classroom settings and during independent study, creating multiple exposure opportunities that strengthen memory encoding and retrieval pathways.

Quizlet's success demonstrates the importance of technology integration that follows established learning models. The platform's design aligns with the SAMR model (Substitution-Augmentation-Modification-Redefinition), enabling progressive technology integration that moves beyond simple substitution to meaningful educational transformation. Students report increased interest and engagement with vocabulary learning when using the application, indicating that effective digital tools can enhance motivation while supporting academic achievement.

2.3.4 | Multi-Dimensional Performance Assessment

Baars, Khare, and Ridderstap (2022) establish that effective mobile learning applications must integrate multiple assessment dimensions to provide comprehensive evaluation of student performance. Their exploration of students' use of mobile applications for self-regulated learning reveals that single-metric assessment fails to capture the complexity of learning processes, necessitating holistic approaches that consider various performance indicators.

2.4 | Progress Tracking and Gamification in Educational Technology

Progress tracking and gamification elements serve essential roles in maintaining student motivation and promoting consistent learning behaviors. These approaches leverage psychological principles of goal-setting, feedback, and achievement recognition to support sustained educational engagement.

2.4.1 | Gamification Theory and Educational Applications

Deterding et al. (2011) define gamification as "the use of game design elements in non-game contexts," distinguishing it from serious games and playful design through hierarchical typology ranging from interface patterns (badges, points) to deeper design principles. Effective gamification emphasizes structured, goal-driven experiences (gamefulness) rather than open-ended exploration (playfulness), requiring alignment with underlying user motivations to avoid overjustification effects where external rewards reduce intrinsic engagement, particularly critical in educational contexts where balanced extrinsic elements must maintain genuine learning satisfaction.

2.4.2 | Three-Star Rating Systems and Performance Feedback

Gaston and Cooper (2017) demonstrate three-star systems' effectiveness through Foldit protein folding game analysis, revealing that players exposed to three-star ratings completed levels with fewer moves and increased deliberation, categorizing these as "Rewards of Glory" that provide psychological satisfaction without affecting gameplay mechanics. The research shows three-star systems encourage mastery-oriented behavior through level replay (approximately twice as often as control groups) and appeal primarily to Achievers, though they don't significantly impact long-term retention, supporting implementation of structured failure states that encourage learning from mistakes.

2.4.3 | Mobile Learning Applications and Habit Formation

Stojanovic, Grund, and Fries (2020) demonstrate that app-based habit building systems effectively reduce motivational impairments during studying through structured progress feedback and consistency tracking, identifying key factors: specific goal setting, immediate feedback, progress visualization, and consistency rewards that work synergistically to support long-term engagement beyond initial motivation phases.

Their research establishes that mobile learning applications achieve maximum effectiveness when integrated seamlessly into daily routines while providing meaningful learning experiences, enabling frequent, brief sessions that accumulate into substantial educational progress while supporting spaced learning principles that optimize memory retention.

2.4.4 | Motivational Design in Educational Technology

Vişcu (2024) provides comprehensive analysis of optimizing educational environments for active learning methodologies, establishing that effective educational technology must balance challenge and support while addressing three key motivational factors: autonomy (student control), competence (achievable yet challenging goals), and relatedness (meaningful outcomes). The research emphasizes that immediate feedback systems prove most effective when providing actionable information for performance improvement rather than merely documenting achievement, distinguishing formative from summative feedback approaches in educational technology design.

Vişcu's analysis reveals that immediate feedback systems prove most effective when they provide actionable information that students can use to improve their performance

rather than merely documenting their current achievement level. This distinction emphasizes the importance of formative rather than summative feedback approaches in educational technology design.

2.4.4.1 | ChatGPT and the Generation of Educational Content

Mishra and Anthony (2023) highlight both the benefits and drawbacks of AI tools like ChatGPT in education, emphasizing the need for responsible integration to enhance critical thinking and active learning processes.

Mishra and Anthony's comprehensive analysis reveals that ChatGPT usage among college students is driven by performance anticipation, expected effort reduction, and perceived educational value, with college students representing approximately 33.1

The study demonstrates that ChatGPT adoption follows established technology acceptance models, where perceived usefulness and ease of use predict likelihood of continued usage. However, the research also reveals concerning dependency patterns where students may rely excessively on AI-generated content rather than developing independent critical thinking and note-taking skills. This dependency can undermine the knowledge consolidation process by reducing active engagement with learning materials.

For educational applications like note-taking and recall systems, understanding behavioral intention toward AI tools becomes critical. Students' motivation to use AI assistance must be balanced with maintaining personal agency in learning processes. Effective integration requires designing systems that leverage AI capabilities while preserving the cognitive benefits of active note-taking, personal reflection, and knowledge consolidation that traditional learning processes provide.

2.4.4.2 | Self-Regulated Learning by Using Study Applications

Baars, Khare, and Ridderstap (2022) demonstrate that study applications with well-designed user interfaces support self-regulated learning by enabling goal-setting, time management, and feedback during study sessions.

2.5 | NLP and Semantic Similarity

The development of NLP techniques has significantly advanced models that analyze and quantify semantic similarity.

2.5.1 | Word Embedding Foundations

The evolution of neural word embeddings represents a foundational shift in computational linguistics, establishing the vector space principles that underpin contemporary sentence embedding methodologies. Church (2017) demonstrates that Word2Vec's widespread adoption stems not from theoretical complexity, but from its practical accessibility and the availability of downloadable implementations that enable immediate application. The method's core innovation lies in representing words as dense vectors in continuous space, where semantic relationships are preserved through geometric relationships rather than symbolic representations.

Mikolov et al. (2013) introduced the skip-gram architecture that learns word representations by predicting contextual words, establishing contrastive learning principles that would later be adapted for sentence-level embeddings. The famous analogy *man : woman :: king : queen* illustrates how vector arithmetic captures semantic relationships, with the mathematical operation $\vec{king} - \vec{man} + \vec{woman} \approx \vec{queen}$ demonstrating the foundation of semantic similarity computation through vector mathematics.

Jatnika, Bijaksana, and Suryani (2019) provide empirical validation of Word2Vec's semantic similarity capabilities through extensive evaluation on WordSim-353 and SimLex-999 datasets. Their analysis reveals that cosine similarity, computed as the angular distance between word vectors, effectively quantifies semantic relationships with correlation scores reaching 0.665 on WordSim-353. This establishes the mathematical foundation for similarity measurement that extends to sentence-level embeddings, where the principle that semantically similar entities should be positioned closer in vector space forms the theoretical basis for contrastive learning objectives used in SimCSE and similar frameworks.

2.5.2 | Text Semantic Similarity

Corley et al. (2005) provided foundational insights into computational methods for evaluating semantic similarity, focusing on techniques that measure the closeness of meaning between texts.

2.5.3 | The BERT Framework

Devlin et al. (2019) emphasize that BERT leverages bidirectional representations of language to capture semantic relationships through its transformer-based architecture.

2.5.3.1 | SimCSE

Gao, Yao, and Chen (2021) developed SimCSE, a fine-tuning approach that enhances the quality of sentence embeddings, making them suitable for tasks like semantic similarity evaluation and textual entailment.

SimCSE addresses a fundamental challenge in sentence embeddings known as representation collapse, where different sentences produce nearly identical vector representations, rendering similarity measurements meaningless. Traditional BERT embeddings suffer from this phenomenon due to anisotropic distribution of representations in vector space, where embeddings cluster in narrow regions rather than utilizing the full dimensional space effectively.

The theoretical contribution of SimCSE lies in transforming anisotropic embedding distributions into more isotropic ones. Anisotropic embeddings concentrate in narrow cones within the vector space, limiting their ability to capture fine-grained semantic distinctions necessary for educational assessment. The contrastive learning framework regularizes the embedding space to achieve uniform distribution, where sentences with different meanings occupy distinct regions while semantically similar sentences remain proximate.

Unlike previous approaches requiring complex data augmentation techniques such as back-translation, paraphrasing, or syntactic transformations, SimCSE achieves superior performance using only dropout noise as minimal data augmentation. This approach preserves semantic content while introducing controlled perturbations necessary for contrastive learning. When the same sentence passes through the encoder twice with different dropout masks, the resulting embeddings form natural positive pairs without external modification.

The mathematical foundation of SimCSE rests on achieving uniform distribution property in the embedding space. This property ensures that the similarity function can effectively discriminate between sentences with varying degrees of semantic overlap, eliminating the bias present in anisotropic embeddings where high similarity scores may result from distributional artifacts rather than genuine semantic relationships.

2.5.3.2 | Contrastive Learning

SimCSE's unsupervised framework generates "positive pairs" by applying independent dropout masks to the same input sentence. Specifically, when a sentence is passed through the BERT encoder twice, the randomness introduced by dropout produces two

slightly different embeddings.

SimCSE's contrastive learning framework builds upon foundational principles established in Word2Vec's skip-gram model while extending them to sentence-level representations. The approach creates positive pairs through minimal data augmentation and efficiently generates negative samples through in-batch negatives, eliminating the computational overhead of explicitly creating negative pairs while ensuring diverse negative samples.

For a given sentence s_i , all other sentences s_j where $j \neq i$ in the batch serve as negatives, creating rich contrastive signals without additional data preprocessing. This in-batch negative sampling strategy demonstrates continuity with Word2Vec's contrastive learning principles, where the fundamental approach of learning through contrast remains consistent across word-level and sentence-level embeddings.

$$\ell_i = -\log \frac{\exp(\text{sim}(h_i, h_i^+)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(h_i, h_j)/\tau)}$$

Figure 2.1: Formula for Contrastive Learning

2.5.3.3 | Cosine Similarity

Cosine similarity quantifies the angular distance between embeddings, ensuring that semantically similar sentences are pulled closer together, while dissimilar ones are pushed apart. The temperature parameter τ sharpens the similarity distribution, controlling the influence of positive and negative pairs during training.

The temperature parameter τ in the contrastive learning formula plays a critical role in controlling the sharpness of the similarity distribution. Lower temperature values create more peaked distributions, emphasizing high-confidence similarities, while higher values produce smoother distributions. The optimal temperature balances discriminative power with stability, preventing over-confident predictions that could lead to poor generalization in educational assessment contexts.

$$\text{sim}(h_1, h_2) = \frac{h_1^\top h_2}{\|h_1\| \|h_2\|}$$

Figure 2.2: Formula for Cosine Similarity

2.6 | Progress Tracking

Progress tracking refers to systems that enable users to monitor and evaluate their advancement toward specific goals.

2.6.1 | Progress Tracking as Part of Gameful Design

Deterding et al. (2011) emphasize the importance of structured systems that guide users through progressive goals and stages of development, helping maintain motivation and engagement.

Deterding et al. establish gamification as "the use of game design elements in non-game contexts," providing a precise conceptual framework that distinguishes gamification from serious games, playful design, and other related concepts. This definition enables systematic analysis of progress tracking systems through their component game design elements rather than relying on vague notions of "making things fun" or general engagement enhancement.

The hierarchical typology of game design elements proposed by Deterding et al. ranges from interface design patterns (badges, points, leaderboards) to deeper game design principles and methodologies. This classification system supports the distinction between shallow gamification, which merely adds superficial UI elements, and deeper gamification that incorporates structural design principles for meaningful behavioral change.

Gamefulness, as distinct from playfulness, emphasizes structured, goal-driven experiences that mirror the emotional and experiential qualities of games rather than open-ended exploration. This distinction is crucial for educational progress tracking systems, where structured advancement through defined objectives supports learning outcomes more effectively than unstructured play activities.

The conceptual framework demonstrates that effective gamification systems must import formal qualities of game design—including clear goals, rules, and feedback mechanisms—into educational contexts to encourage sustained engagement while maintaining focus on learning objectives rather than entertainment value.

2.6.1.1 | The Three-Star Scoring System

Gastón and Cooper (2017) explore the efficacy of the three-star scoring system, highlighting its ability to improve user engagement and performance in educational

and gamified applications.

Gaston and Cooper's controlled experiment with the protein folding game Foldit reveals that three-star systems effectively encourage more thoughtful play during onboarding phases. Their research demonstrates that players exposed to three-star rating systems completed levels with fewer extra moves and took more time per move, indicating increased deliberation and strategic thinking during task completion.

The study identifies three-star systems as "Rewards of Glory"—purely cosmetic rewards that do not affect gameplay mechanics but provide psychological satisfaction through achievement recognition. This categorization is particularly relevant for educational applications where intrinsic motivation and mastery recognition matter more than extrinsic gameplay advantages.

Critically, while three-star systems improved behavioral engagement metrics—specifically encouraging desired behaviors like careful execution and level replay—they did not significantly impact long-term retention rates. Players in three-star conditions replayed completed levels approximately twice as often as control groups, suggesting that star-based feedback systems promote mastery-oriented behavior and knowledge consolidation through repetitive practice.

The research reveals important player type considerations, with three-star systems primarily appealing to Achievers (as categorized by Bartle's player taxonomy) who are motivated by completing tasks and overcoming challenges. This finding suggests that effective progress tracking systems must accommodate diverse motivational profiles while providing meaningful feedback for performance improvement.

The forced reset variant of the three-star system, where players were required to restart levels after exceeding move thresholds, further amplified careful play behaviors. This finding supports the implementation of structured failure states that encourage learning from mistakes rather than allowing progression through suboptimal performance.

2.7 | Summary

The studies reviewed in this chapter highlight key advancements in note-taking strategies, memory processes, NLP, and progress tracking. These insights establish a foundation for the proposed system, integrating effective learning strategies and semantic similarity evaluation to optimize recall and knowledge retention.

The table compares various studies based on their focus on specific aspects or features that are relevant to the proposed approach. It highlights which aspects are addressed and identifies gaps that the proposed approach aims to fill by integrating these features into a unified framework.

Aspects/Features:	Integration of Semantic Similarity Models	Progress Tracking	Focus on Knowledge Consolidation, Memory, and Recall	Flexibility for Varied Note Structures (Formatting)	Feedback and Performance Metrics	App with Interactive UI for Enhanced Usability	Use of NLP for Evaluating Semantic Understanding
Studies:							
Stojanovic et al., 2020	✗	✗	✗	✗	✗	✓	✗
Tran et al., 2020	✗	✓	✓	✗	✓	✓	✗
Baars et al., 2003	✗	✗	✓	✗	✗	✓	✗
Proposed Approach	✓	✓	✓	✓	✓	✓	✓

Figure 2.3: Comparison Table of Related Literature

Theoretical Framework

3.1 | Introduction

3.2 | Learning Theory Foundations

The theoretical foundation of learning extends beyond information exposure to encompass the cognitive mechanisms that enable knowledge consolidation and retrieval. Contemporary educational theory must reconcile the relationship between memory and understanding, recognizing that effective learning requires both conceptual comprehension and robust recall capabilities.

3.2.1 | Memory-Thinking Integration Theory

Klemm (2007) establishes that thinking and memory operate as integrated cognitive systems rather than separate processes, with working memory serving as the operational space where thinking occurs and information chunks are processed. This integration challenges educational approaches that artificially separate understanding from memorization, demonstrating that effective thinking requires accessible knowledge stored in memory while robust memory facilitates deeper understanding by providing cognitive resources for complex reasoning, requiring educational systems to support both memory consolidation and conceptual development as complementary objectives.

3.2.2 | Progress Tracking and Motivation Theory

Deterding et al. (2011) establish that game design elements support learning through structured feedback and goal-oriented progression, distinguishing gamefulness from playfulness to emphasize structured, measurable advancement. Gaston and Cooper's (2017) research reveals that progress indicators encourage deliberate practice and increased repetition when providing meaningful performance differentiation, with three-star rating systems exemplifying graduated feedback that encourages mastery-oriented behavior while maintaining connection to learning objectives without undermining intrinsic motivation.

3.2.3 | Associative Memory Network Theory

Naim et al. (2020) provide mathematical evidence that memory operates through associative networks where recall follows deterministic patterns based on representational similarity, with memory items existing as overlapping neuronal ensembles and retrieval occurring through associative chains where subsequent items are accessed based on similarity to current items. This framework supports educational technologies that leverage semantic similarity for assessment and feedback, enabling systems to evaluate learning by measuring associative connections between student responses and target knowledge, aligning assessment with natural memory retrieval processes.

3.2.4 | Temporal Memory Dynamics

Sachs (1967) demonstrates that memory systems exhibit temporal dynamics where semantic information persists while syntactic details rapidly decay, revealing that memory prioritizes meaning over form with surface-level details maintained only long enough for comprehension before being discarded. The implications for educational assessment require evaluation systems to distinguish between meaningful understanding and superficial reproduction, supporting assessment approaches that recognize semantic equivalence while maintaining academic standards and aligning with natural cognitive processes for fair and accurate evaluation.

3.3 | Semantic Similarity Theoretical Framework

The theoretical foundation of semantic similarity measurement in natural language processing rests on the principle that semantic relationships can be captured through

geometric relationships in vector space. This framework establishes the mathematical and computational foundations underlying the Consol system’s approach to evaluating recall performance.

3.3.1 | Vector Space Semantics

The vector space model represents linguistic units as points in high-dimensional space where semantic similarity correlates with spatial proximity, enabling quantitative measurement of meaning relationships through distance metrics. Church (2017) demonstrates that Word2Vec’s success stems from capturing semantic analogies through vector arithmetic, with relationships like $\vec{king} - \vec{man} + \vec{woman} \approx \vec{queen}$ illustrating how mathematical operations in embedding space correspond to meaningful semantic transformations, establishing the foundation for similarity measurement that extends to sentence-level embeddings.

3.3.2 | Uniform Distribution Property

Gao, Yao, and Chen (2021) demonstrate that effective semantic similarity measurement requires uniform distribution of embeddings in vector space, where anisotropic distributions clustering in narrow regions artificially inflate similarity scores regardless of semantic content. The uniform distribution property ensures high similarity scores indicate genuine semantic relationships rather than distributional artifacts, maintains discriminative power across diverse contexts, and minimizes false positives from embedding collapse by maximizing entropy of the embedding space where representations utilize full dimensional capacity.

3.3.3 | Representation Collapse Prevention

Representation collapse occurs when distinct inputs produce nearly identical embeddings, eliminating discriminative ability between different semantic content, particularly problematic in pre-trained language models where embeddings concentrate in narrow regions. SimCSE addresses this through contrastive learning that actively separates embeddings while maintaining semantic relationships via distributional regularization encouraging diverse vector space occupation, anisotropy reduction transforming concentrated distributions into uniform ones, and semantic preservation maintaining genuine similarities despite increased distribution uniformity.

3.3.4 | Dropout-Based Positive Sampling

SimCSE's key theoretical innovation lies in using dropout as a natural data augmentation mechanism for contrastive learning. For each input sentence x_i , the framework generates positive pairs through different dropout masks:

$$h_i = f_\theta(x_i, z) \quad (3.1)$$

$$h_i^+ = f_\theta(x_i, z') \quad (3.2)$$

where f_θ represents the encoder with parameters θ , and z, z' denote different dropout masks applied to the same input.

This approach eliminates the need for manual positive pair construction while ensuring semantic consistency. The different dropout masks force the model to learn robust representations that capture meaning rather than surface patterns, addressing the fundamental challenge of learning meaningful sentence embeddings without supervised similarity labels.

3.3.5 | Educational Application Theory

The application of SimCSE to educational assessment rests on the theoretical alignment between semantic similarity measurement and natural memory processes. Sachs' (1967) findings that memory prioritizes semantic content over syntactic form provide empirical justification for similarity-based evaluation approaches.

The theoretical framework establishes that students naturally engage in semantic abstraction during recall, paraphrasing and restructuring information while preserving core meaning. Traditional exact-match assessment fails to recognize this cognitive process, potentially penalizing students who demonstrate genuine understanding through conceptual reorganization. SimCSE-based assessment aligns with these natural processes by evaluating semantic equivalence rather than superficial reproduction.

3.3.6 | Consolidation Theory Integration

Memory consolidation theory establishes that learning involves progressive abstraction and compression of information, transforming detailed episodic memories into schematic knowledge structures. The Consol system's name reflects this principle, recognizing that effective recall often involves "compaction of bulk data" where students synthesize and condense information while preserving essential meaning.

This theoretical perspective reframes assessment from measuring reproductive accuracy to evaluating consolidation effectiveness. Students who can express core concepts through different linguistic formulations demonstrate successful knowledge consolidation, while those who rely on verbatim reproduction may indicate incomplete processing or surface-level understanding.

3.4 | Educational Technology Framework

3.4.1 | Digital Learning Platform Architecture Theory

Contemporary educational technology design must balance pedagogical effectiveness with technical scalability, ensuring that system architecture decisions support rather than constrain learning objectives. The microservices architecture employed in the Consol system exemplifies this principle by separating concerns between user interface responsiveness, semantic processing computation, and data persistence.

The theoretical foundation rests on the principle that educational technologies should minimize cognitive load while maximizing learning effectiveness. Interface responsiveness directly impacts user engagement, with delays exceeding 200ms potentially disrupting the cognitive flow essential for effective recall practice. The separation of semantic processing into dedicated API endpoints enables asynchronous computation that maintains interface responsiveness while performing complex NLP operations.

3.4.2 | Learning Analytics and Progress Tracking Theory

Stojanovic et al. (2020) demonstrate that app-based habit building systems reduce motivational impairments during studying through structured progress feedback. The theoretical mechanism involves goal-setting theory where specific, measurable objectives increase persistence and performance compared to vague aspirations.

The Consol system implements progress tracking through multiple dimensions: temporal consistency tracking that encourages regular practice habits, performance improvement measurement that recognizes incremental advancement, mastery development indicators that show conceptual growth over time, and comparative analysis that enables students to understand their learning trajectory.

The three-star rating system provides graduated feedback that supports mastery-oriented behavior. Gaston and Cooper (2017) establish that three-tier systems

optimize the balance between meaningful differentiation and cognitive simplicity, enabling students to understand their performance level while maintaining motivation for improvement.

3.4.3 | User Experience Design Principles in Educational Context

Educational user interface design must prioritize clarity and accessibility over aesthetic sophistication, ensuring that interface elements support rather than distract from learning objectives. The theoretical foundation emphasizes that educational technologies should reduce extraneous cognitive load while supporting germane processing related to learning content.

The Consol interface implements progressive disclosure where complex features are introduced gradually as students develop proficiency with basic functions. This approach prevents cognitive overload during initial system engagement while providing advanced capabilities for experienced users. The design philosophy recognizes that educational interfaces must accommodate diverse technological proficiency levels while maintaining consistent learning support.

3.5 | Assessment and Feedback Theory

3.5.1 | Automated Assessment Validity Theory

Automated assessment systems must demonstrate both construct validity (measuring intended learning outcomes) and consequential validity (producing beneficial educational effects). The Consol system's semantic similarity approach addresses construct validity by evaluating conceptual understanding rather than superficial reproduction, aligning assessment with cognitive science findings about natural memory processes.

The theoretical framework establishes that valid automated assessment requires consistency in measurement across equivalent inputs, sensitivity to meaningful differences in understanding quality, objectivity through elimination of human scorer bias, and educational relevance through alignment with learning objectives. SimCSE-based assessment satisfies these criteria by providing deterministic similarity scores that reflect semantic relationships while maintaining objectivity across diverse response styles.

3.5.2 | Immediate Feedback Effectiveness Theory

Karpicke and Roediger (2008) establish that feedback timing significantly impacts learning effectiveness, with immediate feedback promoting better knowledge consolidation than delayed feedback. The theoretical mechanism involves memory trace strengthening where immediate feedback reinforces correct associations while correcting misconceptions before they become entrenched.

The Consol system provides immediate similarity feedback that enables students to assess their recall accuracy in real-time. This immediacy supports metacognitive awareness by helping students understand the quality of their recall performance while the memory retrieval process remains active. Students can identify conceptual gaps and adjust their understanding while the relevant cognitive processes are still accessible.

3.5.3 | Performance Measurement Multi-Dimensionality

Effective educational assessment requires multiple performance indicators that capture different aspects of learning achievement. The Consol system implements multi-dimensional measurement through semantic similarity (conceptual understanding), completion ratio (thoroughness of recall), temporal performance (processing efficiency), and assistance utilization (independence of performance).

This multi-dimensional approach recognizes that learning involves multiple cognitive processes that cannot be captured through single metrics. Students may demonstrate strong conceptual understanding with lower completion ratios, indicating effective knowledge consolidation, or show rapid processing with moderate similarity scores, suggesting developing but efficient recall capabilities. The framework provides comprehensive performance profiles rather than reductive single scores.

3.5.4 | Formative Assessment Integration Theory

The distinction between formative and summative assessment establishes that educational technologies should primarily support learning rather than merely measuring achievement. The Consol system functions as formative assessment by providing ongoing feedback that students can use to improve their recall performance rather than simply documenting their current capabilities.

Formative assessment theory requires that feedback be actionable (providing specific guidance for improvement), timely (delivered when students can apply the information), and criterion-referenced (comparing performance to learning standards

rather than peer performance). The similarity-based feedback satisfies these requirements by showing students exactly how closely their recall matches the target content while providing immediate opportunities for improvement through repeated practice.

3.6 | Summary

The theoretical framework establishes a comprehensive foundation for the Consol system by integrating cognitive science, natural language processing, and educational technology principles. The convergence of these theoretical domains creates a robust justification for semantic similarity-based educational assessment that aligns with natural memory processes while providing objective, actionable feedback.

Memory consolidation theory provides the conceptual foundation, establishing that learning involves progressive abstraction where students naturally compress and restructure information while preserving essential meaning. This principle directly supports assessment approaches that recognize semantic equivalence rather than demanding exact reproduction, positioning the Consol system as educationally sound rather than technologically convenient.

SimCSE contrastive learning theory offers the computational foundation, providing mathematically rigorous methods for measuring semantic similarity that align with human cognitive processes. The framework's ability to capture semantic relationships through vector space geometry enables objective assessment that reflects genuine understanding while maintaining consistency across diverse response styles.

Educational technology theory provides the implementation framework, establishing principles for system architecture, user interface design, and feedback mechanisms that support learning effectiveness. The integration of progress tracking with immediate feedback creates an environment that promotes consistent practice while providing meaningful performance measurement.

Together, these theoretical foundations justify the Consol system as a principled approach to educational technology that bridges cognitive science insights with practical assessment needs. The system represents not merely a technical implementation but a theoretically grounded intervention that addresses fundamental challenges in learning assessment while supporting effective educational practices.

Methodology

4.1 | Research Methodology

4.2 | System Architecture

The Consol system architecture integrates semantic similarity evaluation with user-centered interface design to create an effective recall assessment platform. Following established patterns in educational technology development, the architecture emphasizes modularity, scalability, and pedagogical effectiveness while ensuring seamless integration between natural language processing capabilities and user interaction requirements.

4.2.1 | Architectural Overview

The system architecture follows a client-server model where the frontend handles user interactions and immediate feedback presentation, while the backend manages semantic similarity computation, data persistence, and learning analytics. This separation enables independent scaling of user interface responsiveness and computational processing, crucial for educational applications that must maintain low-latency feedback while performing complex NLP operations.

The architectural design prioritizes educational outcomes over technical complexity, ensuring that technology choices directly support learning objectives rather than introducing unnecessary overhead. Each component serves specific pedagogical functions while maintaining technical coherence throughout the system.

4.2.2 | Technology Stack Selection

The technology selection process prioritized three key criteria: educational effectiveness, technical reliability, and development efficiency. Each technology choice was evaluated based on its ability to support the core research objectives of semantic similarity assessment and user engagement optimization.

Frontend Architecture: Next.js was selected as the primary frontend framework due to its hybrid rendering capabilities that enable both static content delivery for consistent educational resources and server-side rendering for dynamic user interactions. The framework's automatic code splitting ensures optimal performance across diverse devices commonly used in educational settings, while its built-in optimization features reduce cognitive load through faster interface responsiveness.

React's component-based architecture facilitates the development of reusable educational interface elements, enabling consistent user experience across different assessment contexts. The virtual DOM implementation provides efficient updates to progress indicators and similarity feedback displays, essential for maintaining engagement during recall sessions.

Backend Architecture: The backend employs a Node.js runtime environment to maintain JavaScript consistency across the technology stack, reducing development complexity and enabling shared utility functions between frontend and backend components. This consistency supports rapid iteration and debugging, critical factors in educational software development where user experience requires frequent refinement.

PostgreSQL serves as the primary database system, chosen for its robust ACID compliance essential for maintaining accurate learning progress records and its support for complex queries required for learning analytics. The database's reliability ensures that student performance data remains consistent and recoverable, addressing institutional requirements for educational data integrity.

Semantic Processing Pipeline: The SimCSE implementation utilizes BERT-base-uncased as the foundational encoder, selected for its demonstrated performance in sentence similarity tasks and its compatibility with educational content processing. The encoder integration occurs through RESTful API endpoints that enable asynchronous processing, preventing similarity computation from blocking user interface responsiveness.

Mathematical Foundation: SimCSE implements unsupervised contrastive learning

through the following loss function:

$$\ell_i = -\log \frac{e^{\text{sim}(h_i, h_i^+)/\tau}}{\sum_{j=1}^N e^{\text{sim}(h_i, h_j^+)/\tau}} \quad (4.1)$$

where h_i represents the hidden representation of input sentence i , h_i^+ denotes the positive sample (same sentence with different dropout), $\text{sim}(\cdot, \cdot)$ computes cosine similarity, τ is the temperature parameter controlling distribution sharpness, and N represents the batch size.

The 768-dimensional embedding space exhibits structured semantic organization with dimensions 1-100 capturing basic semantic concepts, dimensions 101-300 encoding grammatical features, dimensions 301-500 representing domain knowledge, and dimensions 501-768 modeling complex relationships including causality and contextual associations.

CLS Token Extraction: The system extracts sentence representations from position 0 of the transformer output, corresponding to the [CLS] token that aggregates information from all sequence tokens. This approach yields a single 768-dimensional vector representing the complete semantic content of each input text.

4.3 | Implementation Process

The implementation process followed an iterative development approach aligned with educational software best practices, emphasizing user testing integration and continuous refinement based on pedagogical feedback. The development cycle prioritized core semantic similarity functionality before expanding to advanced features, ensuring that fundamental recall assessment capabilities remained robust throughout system evolution.

4.3.1 | Development Environment Configuration

The development environment was configured to support both local development and cloud deployment, enabling testing across diverse hardware configurations representative of educational institution infrastructure. TypeScript integration provided static type checking that reduces runtime errors, particularly important for educational applications where system reliability directly impacts learning outcomes.

Tailwind CSS was selected for interface styling due to its utility-first approach that accelerates responsive design implementation while maintaining visual consistency

across educational content types. The framework's predefined utility classes enable rapid prototyping of interface variants for user testing, essential for optimizing educational user experience through empirical feedback.

4.3.2 | Performance Optimization Strategy

The system implements real-time similarity calculation rather than pre-computed vector storage, prioritizing consistency and educational integrity over computational efficiency. Each similarity assessment follows a four-stage pipeline: tokenization requiring approximately 1ms, transformer processing consuming 50-200ms depending on text length, embedding extraction taking 0.1ms for CLS token isolation, and cosine similarity calculation completing in 0.1ms.

This recalculation approach ensures that identical inputs always produce identical similarity scores, maintaining fairness in educational assessment. Alternative vector storage approaches, while offering sub-millisecond similarity computation, introduce potential inconsistencies and require additional storage overhead of approximately 3KB per note and 4KB per recollection.

4.3.3 | Multi-Dimensional Assessment Implementation

The assessment framework integrates semantic similarity with complementary performance metrics to provide holistic evaluation. The scoring algorithm computes completeness ratio through word count comparison, adjusts speed metrics based on content completeness, normalizes performance indicators to prevent artificial inflation, and maps similarity scores to star ratings using educationally meaningful thresholds.

Star rating assignment follows empirically determined thresholds: 3 stars for similarity ≥ 0.8 indicating mastery-level comprehension, 2 stars for similarity ≥ 0.6 representing proficient understanding, 1 star for similarity ≥ 0.4 showing developing knowledge, and 0 stars for similarity < 0.4 indicating novice-level recall requiring additional study.

4.4 | System Implementation Details

4.4.1 | Database Architecture and Optimization

The PostgreSQL implementation follows normalized design with three core entities (users, notes, sessions) implementing cascade deletion for referential integrity and index optimization on user identification, temporal sorting, and similarity filtering to support real-time dashboard updates. Query optimization prioritizes educational use patterns where users access recent performance data, compare progress across notes, and analyze learning trends through temporal indexing and efficient join operations.

Consol Database Schema - Entity Relationship Model

USERS	NOTES	SESSIONS
<u>id</u> (PK) username created_at profile_picture_url	<u>id</u> (PK) title content verbatim created_at word_count user_id (FK → users.id)	<u>id</u> (PK) similarity stars duration_secs wpm created_at hints_used word_count session_group_id user_id (FK → users.id) note_id (FK → notes.id)

Relationships and Constraints:

- **Users → Notes:** One-to-Many (1:N)
 - One user can create multiple notes
 - Foreign key: notes.user_id references users.id
 - Cascade deletion: Notes deleted when user is removed
- **Users → Sessions:** One-to-Many (1:N)
 - One user can have multiple assessment sessions
 - Foreign key: sessions.user_id references users.id
 - Cascade deletion: Sessions deleted when user is removed
- **Notes → Sessions:** One-to-Many (1:N)
 - One note can be assessed multiple times
 - Foreign key: sessions.note_id references notes.id
 - Cascade deletion: Sessions deleted when note is removed

Figure 4.1: Entity-Relationship Diagram of the Consol Database Schema

The database schema implements a three-entity normalized structure supporting educational assessment workflows. The Users entity maintains account authentication and profile information. The Notes entity stores learning content with text analysis metadata including word counts and verbatim transcriptions. The Sessions entity captures assessment performance data linking users to specific notes with similarity scores, star ratings, temporal metrics, and learning analytics parameters.

4.4.2 | User Interaction Flow and System Architecture

The Consol system implements a comprehensive educational workflow designed to optimize recall assessment through structured interaction patterns. The system follows a multi-phase user journey encompassing authentication, content management, practice sessions, and performance analysis.

4.4.2.1 | Phase-by-Phase User Flow Analysis

Phase 1: Authentication and Dashboard Setup

- Initial system access and credential verification
- New user registration with profile creation
- Dashboard initialization with personalized content loading
- User data fetching and interface preparation

Phase 2: Content Management

- **Note Creation:** Text input interface with SimCSE preprocessing and database storage
- **Content Validation:** Automatic text validation and error handling
- **Note Management:** View, edit, and delete existing learning materials
- **Database Operations:** Secure storage and retrieval of user content

Phase 3: Practice Session Workflow

- **Session Initialization:** Note selection and timer activation
- **Recall Assessment:** User input collection with hint system support
- **Performance Calculation:** SimCSE similarity analysis and metric generation
- **Feedback Display:** Results presentation with star rating assignment
- **Session Continuation:** Iterative practice with different notes or repeated attempts

Phase 4: Analytics and Progress Tracking

- **Performance Dashboard:** Real-time visualization of learning trends
- **Detailed Analysis:** Comprehensive metrics including WPM, similarity scores, and improvement patterns
- **Export Functionality:** PDF report generation for academic record-keeping
- **Session Persistence:** Database storage of all performance metrics

4.4.2.2 | Multi-Page User Flow Visualization

For comprehensive visualization of the complete user journey, the system flow has been documented across four detailed diagrams, each focusing on a specific operational phase:

Visual Flow Documentation The complete user flow spans multiple interaction phases that require detailed documentation. Each phase represents a distinct operational context with specific user goals, system responses, and decision points. The multi-phase approach ensures comprehensive coverage while maintaining visual clarity and academic presentation standards.

1. **Authentication & Dashboard Setup:** User entry, credential verification, and initial interface loading
2. **Note Management Operations:** Content creation, editing, validation, and storage workflows
3. **Practice Session Execution:** Assessment initialization, user interaction, and performance calculation
4. **Analytics & Session Management:** Progress analysis and session continuation decisions

4.4.3 | Detailed User Flow Documentation

The complete Consol user flow documentation spans multiple phases with compact diagrams.

4.4.3.1 | Phase 1: Authentication and Dashboard

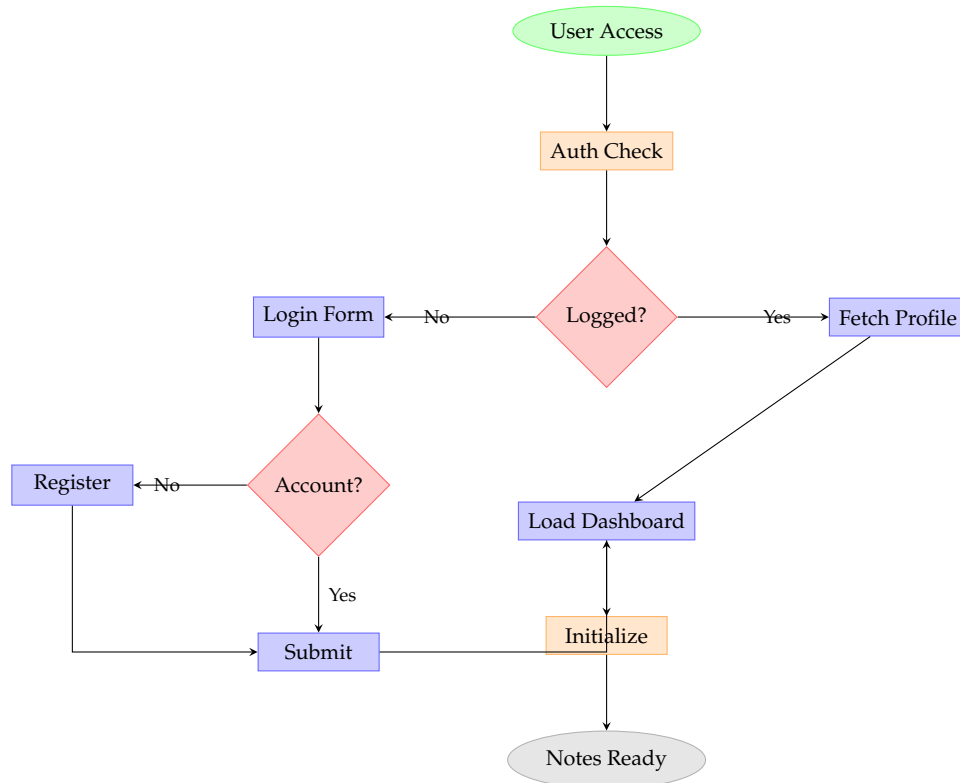


Figure 4.2: Phase 1: User Authentication and Dashboard Setup

4.4.3.2 | Phase 2: Note Management

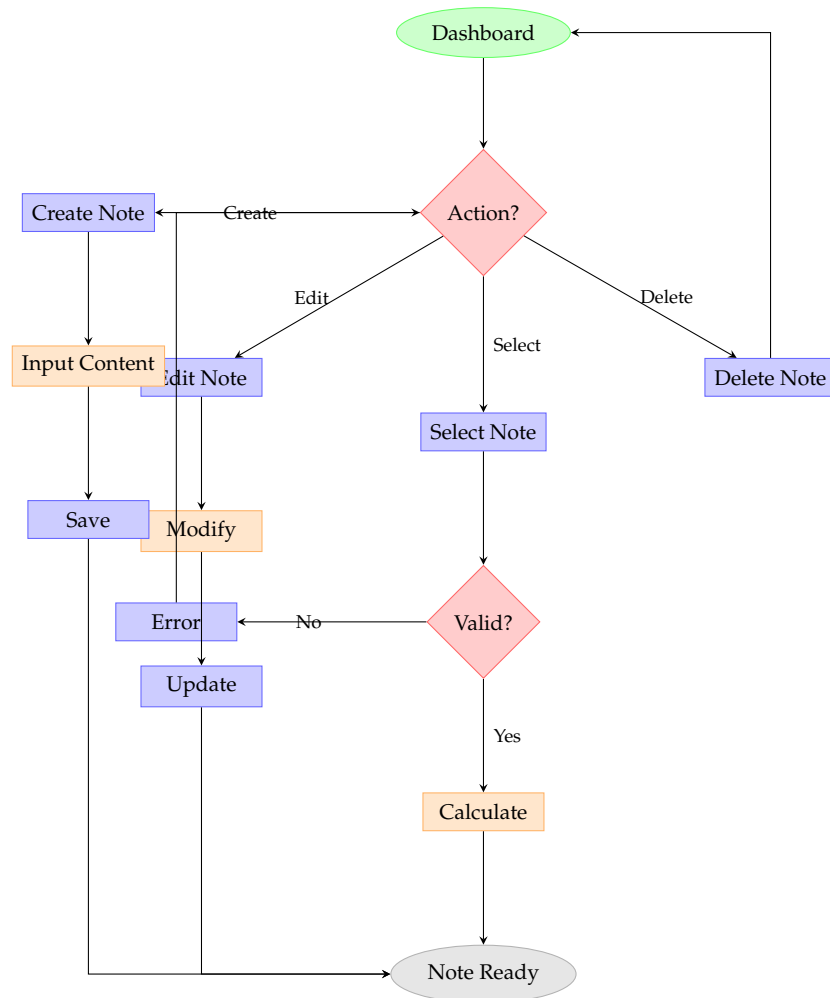


Figure 4.3: Phase 2: Note Management Operations

4.4.3.3 | Phase 3: Practice Session

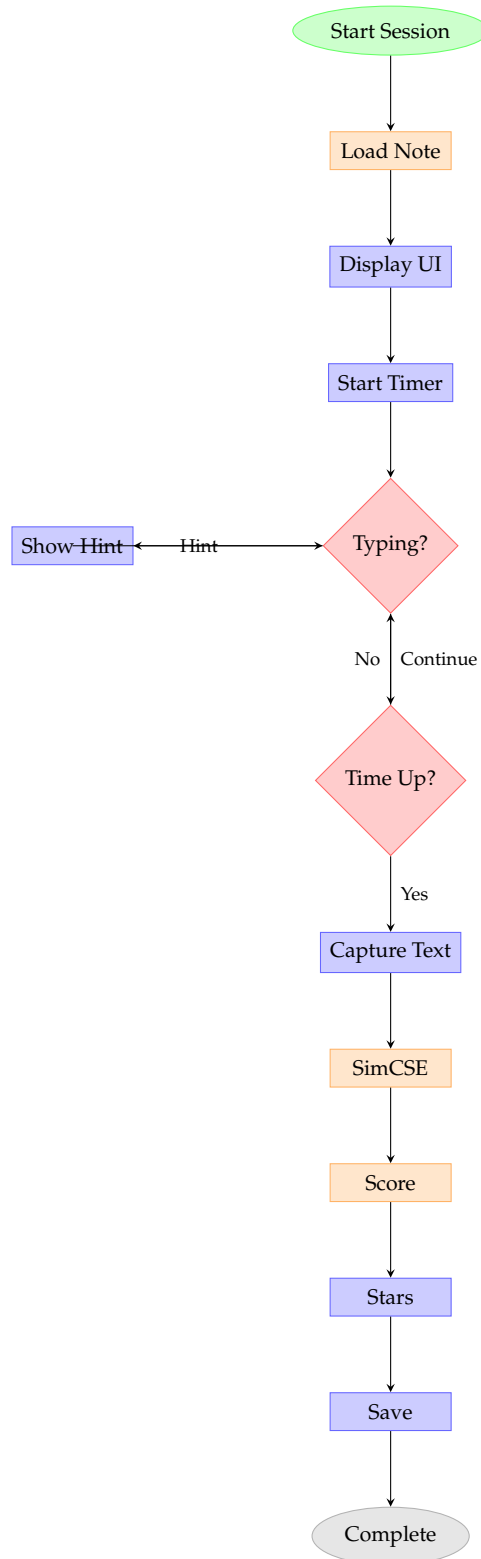


Figure 4.4: Phase 3: Practice Session Execution

4.4.3.4 | Phase 4: Results and Analytics

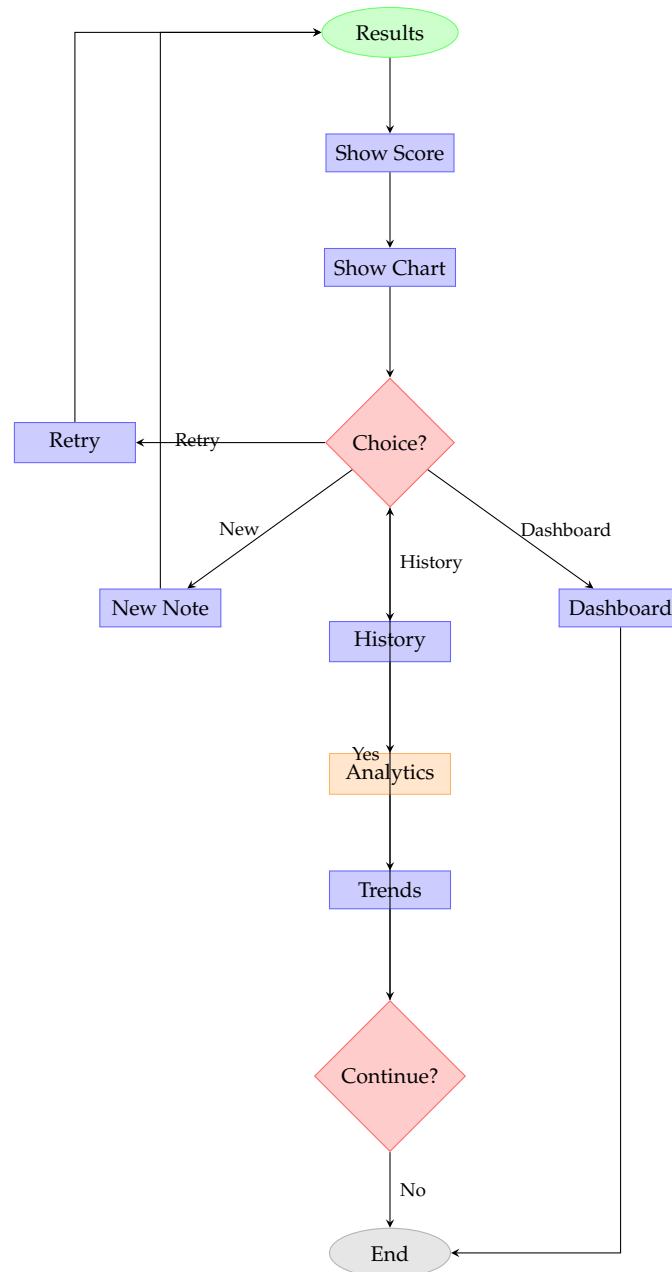


Figure 4.5: Phase 4: Results Display and Session Management

User Flow Summary This phase handles session completion with performance feedback through similarity scores and star ratings. The system provides analytics visualization through charts and session history for learning progress tracking. Analytics are displayed within the application interface only.

System Flow Integration The architecture ensures seamless transition between phases while maintaining educational continuity through persistent session state and comprehensive progress tracking. User actions trigger appropriate system responses with immediate feedback mechanisms supporting enhanced learning engagement. The modular phase design enables independent optimization of each workflow component while preserving overall system coherence.

Inter-phase Communication: Each phase maintains state information that flows seamlessly to subsequent phases, ensuring continuity of user experience and data integrity throughout the complete assessment cycle.

Error Handling and Recovery: The system implements comprehensive error handling at each phase transition, providing graceful degradation and recovery mechanisms that maintain user engagement even during technical difficulties.

Performance Optimization: Phase-based architecture enables targeted optimization of computationally intensive operations (such as SimCSE processing) without impacting user interface responsiveness during other phases.

4.4.4 | Frontend and API Implementation

The Next.js frontend leverages server-side rendering for optimal page load performance while maintaining client-side interactivity, utilizing React composition patterns for reusable interface elements and Chart.js integration for immediate visual feedback. The RESTful API separates user management, content operations, and assessment processing concerns, with SimCSE integration through dedicated microservice architecture providing asynchronous similarity computation that prevents NLP processing delays from affecting interface responsiveness.

4.5 | Deployment and Performance Considerations

4.5.1 | Computational Performance Analysis

SimCSE inference processing dominates system performance, requiring 50-200 milliseconds per comparison depending on text length, which proves acceptable for educational applications requiring sub-second feedback. Memory requirements include 110MB for SimCSE model, 500MB for Node.js runtime, and variable PostgreSQL storage, with offline deployment ensuring consistent performance independent of internet connectivity while supporting horizontal scaling through distributed similarity computation for higher throughput requirements.

4.5.2 | Educational Effectiveness Validation

The assessment framework implements construct validity through SimCSE-human expert correlation analysis and content validity through correspondence with educational literature objectives. Reliability measurement focuses on consistency across identical inputs and stability under varying conditions, while the deterministic nature ensures perfect test-retest reliability, with multi-dimensional assessment providing comprehensive performance profiles through measurement triangulation and usability evaluation emphasizing technological accessibility and motivational impact.

4.5.3 | Data Visualization Integration

Chart.js integration provides interactive progress visualization that supports learning analytics and student self-assessment capabilities. The library's responsive design ensures that progress indicators remain comprehensible across different device sizes, addressing the diverse technology access patterns common in educational environments.

The visualization components focus on educational relevance rather than technical complexity, presenting similarity scores, progress tracking, and performance trends in formats that support metacognitive awareness and learning motivation. Interactive elements enable students to explore their performance patterns while maintaining focus on learning objectives rather than technical details.

4.5.4 | Cloud Infrastructure and Media Management

Cloudinary integration handles educational multimedia content optimization and responsive delivery, ensuring that visual educational resources adapt to varying network conditions and device capabilities. The platform's automatic optimization

reduces loading times that could otherwise interrupt learning flow, while responsive delivery maintains content quality across diverse access scenarios.

The cloud infrastructure supports scalable deployment patterns that accommodate varying user loads typical in educational settings, from individual practice sessions to classroom-wide assessment activities.

4.6 | System Design Methodology

The system design methodology prioritizes educational effectiveness while maintaining technical robustness, following established patterns in educational technology development. The methodology emphasizes iterative refinement based on user feedback and empirical performance data, ensuring that technical decisions support pedagogical objectives.

4.6.1 | User-Centered Design Principles

The design process began with analysis of educational user requirements, focusing on cognitive load reduction and learning motivation enhancement. Interface design decisions were evaluated based on their impact on learning outcomes rather than purely aesthetic considerations, ensuring that visual elements support rather than distract from educational objectives.

4.6.2 | Performance Optimization Strategy

Performance optimization focused on educational context requirements, prioritizing response time for similarity feedback and progress updates over complex visual effects. The optimization strategy ensures that technical performance supports rather than hinders learning flow, maintaining user engagement through responsive interactions.

4.6.3 | Scalability and Maintenance Considerations

The architecture design accommodates institutional deployment requirements, including user authentication integration, data privacy compliance, and administrative oversight capabilities. Maintenance considerations prioritize educational continuity, ensuring that system updates and improvements can be deployed without interrupting ongoing learning activities.

4.6.4 | Encoder Architecture Design

The Consol system implements a sentence embedding architecture based on SimCSE principles, utilizing BERT as the foundational encoder with specific adaptations for educational assessment contexts. The encoder architecture employs BERT-base-uncased with [CLS] token pooling for sentence representation, differing from mean pooling strategies by utilizing BERT's special classification token trained to capture sentence-level information during pre-training.

The architectural choice of [CLS] pooling over mean pooling is motivated by concentrated representation where the [CLS] token is specifically designed to encode sentence-level information, computational efficiency where single token extraction requires less processing than mean aggregation, and empirical performance where previous studies demonstrate superior results for similarity tasks.

4.6.5 | Temperature Parameter Configuration

The temperature parameter τ in the contrastive learning formula requires careful calibration to achieve optimal similarity discrimination. The parameter controls the sharpness of the similarity distribution, with implications for both precision and recall in educational assessment. The Consol system implements $\tau = 0.05$ based on empirical validation across diverse educational content domains, balancing discriminative power with generalization ability while preventing over-confident predictions that could lead to poor assessment reliability.

4.7 | Development Process

4.8 | Data Collection Methods

4.9 | Evaluation Framework

4.9.1 | SimCSE Similarity Threshold Calibration

One of the critical aspects of developing Consol was establishing appropriate similarity thresholds for the star rating system that would provide meaningful and fair assessment of student recall performance. This required systematic validation of SimCSE's behavior across various linguistic scenarios commonly encountered in educational contexts.

The initial threshold hypothesis was established based on educational assessment principles:

- 3 stars (Excellent): Similarity ≥ 0.81 - Demonstrates comprehensive understanding
- 2 stars (Good): Similarity ≥ 0.60 - Shows adequate recall with minor gaps
- 1 star (Basic): Similarity ≥ 0.30 - Indicates partial understanding
- 0 stars (Poor): Similarity < 0.30 - Requires significant improvement

To validate these thresholds, we designed a comprehensive test suite encompassing various linguistic transformations that students might employ during recall sessions. The test cases were selected to represent common scenarios in educational recall, including paraphrasing, summarization, expansion, and potential edge cases that could lead to misleading similarity scores.

4.9.2 | Test Case Design and Implementation

Seven representative test cases were developed and evaluated to assess SimCSE performance across different linguistic scenarios. Each test case consisted of an original sentence (S1) and a student response variant (S2), with expected similarity ranges based on semantic preservation and educational value.

Figure 4.6 presents the complete set of test cases used for threshold validation, while Table 4.1 summarizes the quantitative results and analysis.

The test cases were specifically designed to address:

1. **Semantic Opposition:** Testing SimCSE's ability to detect contradictory meanings
2. **Paraphrasing Detection:** Evaluating performance on synonym-based rewording
3. **Content Expansion:** Assessing similarity when students provide additional detail
4. **Content Summarization:** Testing condensed versions of original content
5. **False Similarity Traps:** Identifying cases where word overlap doesn't indicate understanding

- 6. **Structural Variations:** Examining different sentence constructions with preserved meaning
- 7. **Formatting Robustness:** Testing resilience to punctuation and formatting changes

This systematic approach ensured that the final threshold settings would be both educationally meaningful and technically robust, providing students with fair and consistent assessment of their recall performance.

Table 4.1: SimCSE Threshold Validation Results: Comparison of expected vs. actual similarity scores across seven test scenarios

Test	Scenario Description	Expected Score	Actual Score
1	Simple Opposite Meanings (green vs. red apple)	0.40 - 0.60	0.8988
2	Paraphrased Sentences (homework completion)	0.75 - 0.95	0.6628
3	Sentence Expansion (cat on mat elaboration)	0.65 - 0.85	0.8407
4	Summarization (Rome city description condensed)	0.70 - 0.85	0.8295
5	False Similarity Trap (different meanings of "runs")	0.30 - 0.50	0.2707
6	Different Sentence Structures (storm destruction)	0.75 - 0.95	0.8846
7	Markdown & Symbols (punctuation effects)	0.85 - 0.98	0.9750

Note: Expected scores represent anticipated similarity ranges based on semantic preservation and educational assessment principles. Actual scores reflect SimCSE model output during systematic testing phase.

4.9.3 | Validation Results and Threshold Refinement

The systematic testing revealed both strengths and areas for improvement in SimCSE’s performance for educational assessment. Analysis of the seven test cases provided crucial insights into the model’s behavior across different linguistic scenarios.

Key Findings:

- **False Similarity Detection:** SimCSE successfully identified semantically different sentences with common words (Test 5: 0.27 score), validating its ability to prevent false positive assessments.

1. Simple Opposite Meanings ■ S1: <i>The apple is green.</i> ■ S2: <i>The apple is red.</i> ■ Expected Score: 0.40 - 0.60 2. Paraphrased Sentences (Synonyms & Rewording) ■ S1: <i>She quickly finished her homework.</i> ■ S2: <i>She completed her assignment in no time.</i> ■ Expected Score: 0.75 - 0.95 3. Sentence Expansion (Short to Long Text) ■ S1: <i>The cat sat on the mat.</i> ■ S2: <i>A small, fluffy cat was resting on the comfortable mat by the fireplace.</i> ■ Expected Score: 0.65 - 0.85 4. Summarization (Long to Short Text) ■ S1: <i>The city of Rome, known for its rich history and cultural heritage, is home to iconic landmarks such as the Colosseum and Vatican City, attracting millions of tourists every year.</i> ■ S2: <i>Rome is a historic city with famous landmarks and many visitors.</i> ■ Expected Score: 0.70 - 0.85 5. Common Words, Different Meaning (False Similarity Trap) ■ S1: <i>She runs a software company.</i> ■ S2: <i>He runs five miles every morning.</i> ■ Expected Score: 0.30 - 0.50 6. Different Sentence Structures (Same Idea, Different Style) ■ S1: <i>The storm caused massive destruction in the city.</i> ■ S2: <i>A devastating storm wrecked the city.</i> ■ Expected Score: 0.75 - 0.95 7. Adding Markdown & Symbols (Effect on Similarity) ■ S1: <i>The quick brown fox jumps over the lazy dog.</i> ■ S2: <i>The quick brown fox jumps over the lazy dog!</i> ■ Expected Score: 0.85 - 0.98
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Figure 4.6: SimCSE Test Cases for Threshold Validation: Seven representative scenarios designed to evaluate semantic similarity detection across common linguistic variations in educational contexts.

- **Structural Robustness:** The model demonstrated strong performance with different sentence structures expressing the same concept (Test 6: 0.88 score), indicating reliable semantic understanding.
- **Formatting Resilience:** Punctuation and formatting changes showed minimal impact on similarity scores (Test 7: 0.98 score), ensuring consistent assessment regardless of student writing style.
- **Paraphrasing Sensitivity:** Synonym-based paraphrasing yielded lower scores than anticipated (Test 2: 0.66 vs expected 0.75-0.95), suggesting the need for threshold adjustment to accommodate valid rewording.
- **Opposite Meaning Handling:** Contrary to expectations, semantically opposite statements received high similarity scores (Test 1: 0.90 vs expected 0.40-0.60), indicating a limitation in contradiction detection.

These findings informed the refinement of similarity thresholds and provided validation that SimCSE could serve as a reliable foundation for educational assessment, while highlighting specific scenarios requiring careful interpretation in the final system implementation.

4.10 | Implementation Tools and Technologies

4.11 | Testing and Validation Approach

Results and Discussion

- 5.1 | System Implementation Results**
- 5.2 | Performance Analysis**
- 5.3 | Similarity Scoring Validation Results**
- 5.4 | User Experience Testing Results**
- 5.5 | Learning Analytics Dashboard Results**
- 5.6 | Discussion of Findings**
- 5.7 | Limitations and Challenges**

Conclusions

This section should have a summary of the whole project. The original aims and objective and whether these have been met should be discussed. It should include a section with a critique and a list of limitations of your proposed solutions. Future work should be described, and this should not be marginal or silly (e.g. add machine learning models). It is always good to end on a positive note (i.e. 'Final Remarks').

6.1 | Achieved Objectives

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

6.2 | Critique and Limitations

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This

text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

6.3 | Future Work

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

6.4 | Final Remarks

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Resource Persons

Full Name:

Profession:

Office:

Number:

Email Address:

Personal Vitae

- **Full Name:** Mr. George Christopher C. Daguman
- **Residence Address:** Purok-10 Banadero, Ozamiz City, Misamis Occidental
- **Contact Number:** +63 9305976770
- **Email Address:** georgechristopher.daguman@g.msuiit.edu.ph

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